- Marginal Revenue is interpreted as the approximate change in revenue made on by selling of $(x+1)^{th}$ unit, when the sale level is x units.
- A function $f(x)$ is said to be increasing function in the interval $[a, b]$ if $x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$ for all $x_1, x_2 \in [a, b]$
- A function $f(x)$ is said to be strictly increasing in $[a, b]$ if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$ for all $x_1, x_2 \in [a, b]$
- A function $f(x)$ is said to be decreasing function in $[a, b]$ if $x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)$ for all $x_1, x_2 \in [a, b]$
- A function $f(x)$ is said to be strictly decreasing function in $[a, b]$if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$ for all $x_1, x_2 \in [a, b]$
- Let f be a differentiable function on an open interval (a, b) containing c and suppose that $f'(c)$ exists.

  (i) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.

  (ii) If $f'(c) = 0$ and $f''(c) < 0$,then f has a local maximum at c.
- A function $f(x, y)$ of two independent variables x and y is said to be homogeneous in x and y of degree n if for $t > 0$ $f(tx, ty) = t^n f(x, y)$
- If $u = f(x, y)$ is a homogeneous function of degree n, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$
- The partial elasticity of demand q with respect to p_1 is defined to be $\eta_{qp_1} = \frac{Eq}{Ep_1} = \frac{-p_1}{q}\frac{\partial q}{\partial p_1}$
- The partial elasticity of demand q with respect to p_2 is defined to be $\eta_{qp_2} = \frac{Eq}{Ep_2} = \frac{-p_2}{q}\frac{\partial q}{\partial p_2}$

Some standard results

1. Total cost: $C(x) = f(x) + k$
2. Average cost: $AC = \frac{f(x)+k}{x} = \frac{c(x)}{x}$
3. Average variable cost: AVC$= \frac{f(x)}{x}$
4. Average fixed cost: $AFC = \frac{k}{x}$
5. Marginal cost: $MC = \frac{dC}{dx}$
6. Marginal Average cost: $MAC = \frac{d}{dx}(AC)$
7. Total cost: $C(x) = AC \times x$
8. Revenue: $R = px$
9. Average Revenue: $AR = \frac{R}{x} = p$

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